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Body of Revolution High Alpha Aerodynamics

25 July 2013

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Introduction

When rockets with recovery requirements are flown to higher and higher altitudes the point is reached where familiar techniques no longer work. The problem is high dynamic pressure during descent. Even if a drogue is deployed near apogee the dynamic pressure build up as the denser lower altitudes are reached will exceed the strength of any drogue. Later drogue deployment would only exacerbate this problem.

The classical solution to this problem is to separate the payload from the rocket body, or the fins from everything else, trim the recovery package to ensure a high angle of attack (alpha), and then let aerodynamic drag slow it to modest subsonic speed before deploying a drogue parachute. Fin removal is essential to preclude lawn dart behavior.

The high angle of attack aerodynamics of a slender body of revolution documented here are coded into JORGENSN.xls.

Mnemonics

Mnemonic	Definition
d_{REF}	Reference length upon which the aerodynamic coefficients are based, ft
S_{REF}	Reference area upon which the aerodynamic coefficients are based, ft^2
M_∞	Free stream Mach number
M_N	Cross flow Mach number $=M_\infty \sin\alpha$
C_{DN}	Cross flow drag coefficient based on crossflow dynamic pressure and local body radius
X_{CG}	Body station location of the center of gravity, in aft of the nose tip
C_A	Axial force coefficient
C_{A0}	Axial force coefficient at zero degrees angle of attack
C_{A180}	Axial force coefficient at 180 degrees angle of attack
C_N	Normal force coefficient
$C_{N\alpha}$	Normal force coefficient slope
$C_{N\alpha 0}$	Normal force coefficient slope at zero degrees angle of attack
$C_{N\alpha 180}$	Normal force coefficient slope at 180 degrees angle of attack

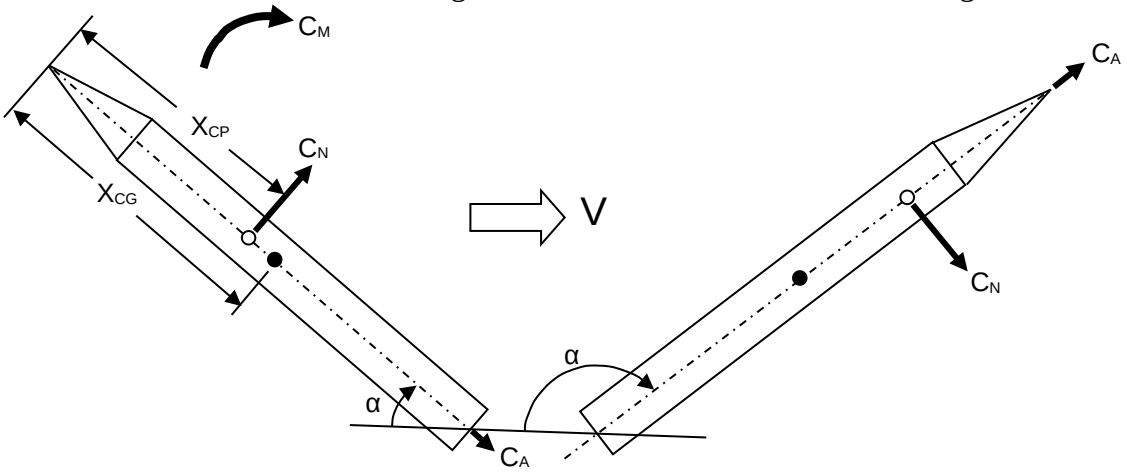
Mnemonic	Definition
X_{CP0}	Body station location of the center of pressure at zero

	degrees angle of attack, in aft of the nose tip
X_{CP90}	Body station location of the center of pressure at 90 degrees angle of attack, in aft of the nose tip
X_{CP180}	Body station location of the center of pressure at 180 degrees angle of attack, in aft of the nose tip
R_{ED}	Reynolds number based on ambient fluid properties and d_{REF}
R_{EN}	Crossflow Reynolds number based on D_{REF}
R_i	ith tabular body radius at station X_i, in
X_i	ith body station, in aft of the nose tip
L_{OA}	Overall length of the body, ft
L_i	Length of the ith body segment, ft
AR	Body planform aspect ratio,
A_{BROAD}	Broadside projected area of the body, ft^2
S_{BROAD}	Axial projected area of the body, ft^2
D_{AVE}	Average body diameter, ft
V	Flight speed, ft/sec
α	Angle of attack, rad
η	Ratio of crossflow drag for a finite-length cylinder to that for an infinite-length cylinder

Analysis

This memo documents the process used to estimate the high angle of attack aerodynamics of a body of revolution needed to support this kind of recovery scheme. For the most part, this follows ref. (1). The aerodynamics at alpha = 0 and 180 degrees are assumed to be generated off line in other codes. There is no offline code to generate the aerodynamics at alpha = 90 degrees so this data is generated internally.

After estimating the aerodynamics as a function of angle of attack, JORGENSEN.xls finds the trim angle of attack and the associated trim drag coefficient.



In the sketch above, the two angle of attack regimes are displayed. First, consider $0^\circ \le \alpha \le 90^\circ$. Since this is an interpolative scheme,

$$C_A = C_{A0}\cos^2(\alpha) + C_{A90}\sin^2(\alpha),$$

(1)

$$C_N = (1/2)C_{N\alpha 0}\sin(2\alpha)\cos(a/2) + \eta C_{DN}A_{BROAD}\sin^2(\alpha)/S_{REF},$$

(2)

$$C_{N\alpha} = (1/2)C_{N\alpha 0}\left[2\cos(2\alpha)\cos(\alpha/2) - (1/2)\sin(2\alpha)\sin(\alpha/2)\right] + 2\eta C_{DN}A_{BROAD}\sin(\alpha)\cos(\alpha)/S_{REF},$$

(3)

$$C_M = \left[C_{N\alpha 0}\left(\frac{X_{CG} - X_{CP0}}{2d_{REF}}\right)\right]\sin(2\alpha)\cos(\alpha/2) + \eta C_{DN}A_{BROAD}\left(\frac{X_{CG} - X_{CP90}}{S_{REF}d_{REF}}\right)\sin^2(\alpha)$$

(4)

and

$$C_{M\alpha} = \left[C_{N\alpha 0}\left(\frac{X_{CG} - X_{CP0}}{2d_{REF}}\right)\right]\left[2\cos(2\alpha)\cos(\alpha/2) - (1/2)\sin(2\alpha)\sin(\alpha/2)\right] + 2\eta C_{DN}A_{BROAD}\left[\frac{X_{CG} - X_{CP90}}{S_{REF}d_{REF}}\right]\sin(\alpha)\cos(\alpha).$$

(5)

Before proceeding to the case of $90^\circ \le \alpha \le 180^\circ$ a discussion of how one might estimate the parameters in these equations is needed. First, C_{A0} can be found from ref. (4). $C_{N\alpha 0}$ and X_{CP0} can be estimated from the codes in ref. (3). There are two of these, one for subsonic flight and the other for supersonic flight. The remaining aerodynamic parameters are for $\alpha = 90^\circ$ and will be discussed later. The non-aerodynamic parameters, S_{REF}, d_{REF} and X_{CG} address the configuration under study.

For angles of attack > 90° the sketch above shows that the analysis is like that above with the ends swapped:

$$C_A = -C_{A180}\cos^2(\alpha) + C_{A90}\sin^2(\alpha),$$

(6)

$$C_N = -(1/2)C_{N\alpha 180}\sin(2\alpha)\sin(a/2) + \eta C_{DN}A_{BROAD}\sin^2(\alpha)/S_{REF},$$

(7)

$$C_{N\alpha} = -(1/2)C_{N\alpha 180}\left[2\cos(2\alpha)\sin(\alpha/2) + (1/2)\sin(2\alpha)\cos(\alpha/2)\right] + 2\eta C_{DN}A_{BROAD}\sin(\alpha)\cos(\alpha)/S_{REF}, \text{ and}$$

(8)

$$C_M = \left[C_{N\alpha 180} \left(\frac{X_{CG} - X_{CP180}}{2d_{REF}} \right) \right] \sin(2\alpha)\sin(\alpha/2) + \eta C_{DN} A_{BROAD} \left(\frac{X_{CG} - X_{CP90}}{S_{REF} d_{REF}} \right) \sin^2(\alpha), \tag{9}$$

$$\begin{aligned} C_{M\alpha} = & \left[C_{N\alpha 180} \left(\frac{X_{CG} - X_{CP180}}{2d_{REF}} \right) \right] \left[2\cos(2\alpha)\sin(\alpha/2) + (1/2)\sin(2\alpha)\cos(\alpha/2) \right] \\ & + 2\eta C_{DN} A_{BROAD} \left[\frac{X_{CG} - X_{CP90}}{S_{REF} d_{REF}} \right] \sin(\alpha)\cos(\alpha). \end{aligned} \tag{10}$$

Data for parameter estimation can be obtained as described above, except for the case of a blunt, flat-faced nose for which recourse to ref. (4) is needed. While the drag for a blunt, flat-faced nose is easy to estimate from ref. (4), the associated normal force coefficient is not. First, by definition, the flat face itself will not carry normal force near $\alpha = 180°$. The question is, how much normal force is carried over to the cylindrical afterbody? To answer this, consider a blunt cone-cylinder. In subsonic flow, there is no cylinder normal force. The same is true for hypersonic (Newtonian) flow. In the limit as the cone approaches a flat face, this is expected to remain true. Thus, in practice, assume zero $C_{N\alpha 180}$ for a flat face.

Finally, in general

$$C_D = C_A \cos(\alpha) + C_N \sin(\alpha) \tag{11}$$

The conditions for static equilibrium, or trim, are that

$$\begin{aligned} C_M &= 0 \\ C_{M\alpha} &< 0 \end{aligned} \tag{12}$$

Broadside Aerodynamics

Start by defining the properties associated to the body profile:

$$\begin{aligned} \delta &= \tan^{-1}\left(\frac{R_{i+1} - R_i}{X_{i+1} - X_i} \right), \\ L_i &= (X_{i+1} - X_i)/12, \end{aligned} \tag{13}$$

Next,

$$S_{BROAD} = (1/24)\sum L_i (R_{i+1} + R_i)\cos(\delta_i)\sin(\delta_i), \text{ and}$$

$$A_{BROAD} = (1/24)\sum L_i (R_{i+1} + R_i)\cos^2(\delta_i)\,. \tag{14}$$

$$SumXS = (1/36)\sum L_i^2 (R_i + 2R_{i+1})\cos^2(\delta_i)\,. \tag{15}$$

With these, the broadside center of pressure can be found:

$$X_{CP90} = SumXS / A_{BROAD}\,. \tag{16}$$

Next,

$$D_{AVE} = A_{BROAD} / L_{OA}\text{, and} \tag{17}$$

$$AR = L_{OA}{}^2 / A_{BROAD} \tag{18}$$

$$M_N = M_\infty \sin(\alpha)\text{, and}$$

$$R_{EN} = R_{ED} D_{AVE} \sin(\alpha) / d_{REF}\,. \tag{19}$$

The broadside drag coefficient C_{DN} and its aspect ration correction factor η are both represented as the product of two functions:

$$C_{DN} = f_1(M_N) * f_2(R_{EN})\text{, and} \tag{20}$$

Note that if $M_N > 0.45$, then $f_2 = 1$

$$\eta = f_3(AR) * f_4(M_N)\,. \tag{21}$$

These functions are not known as analytical functions, but only in terms of their tabular values as derived from experiment in ref. (1)

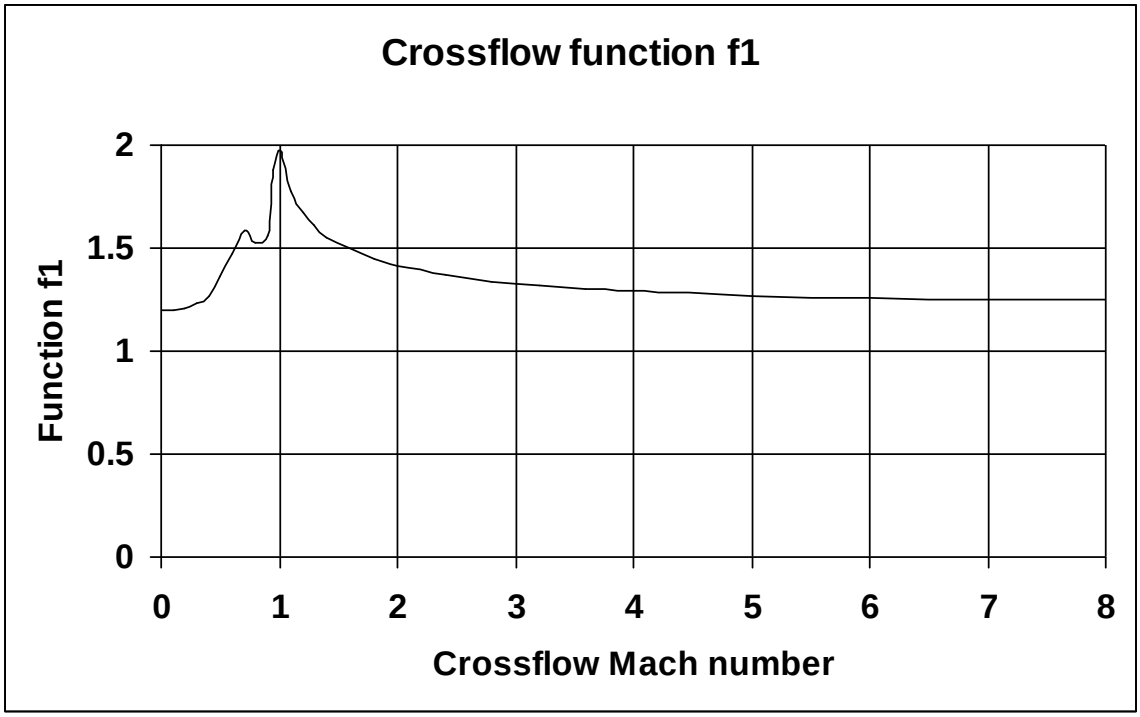
With these results, we find that

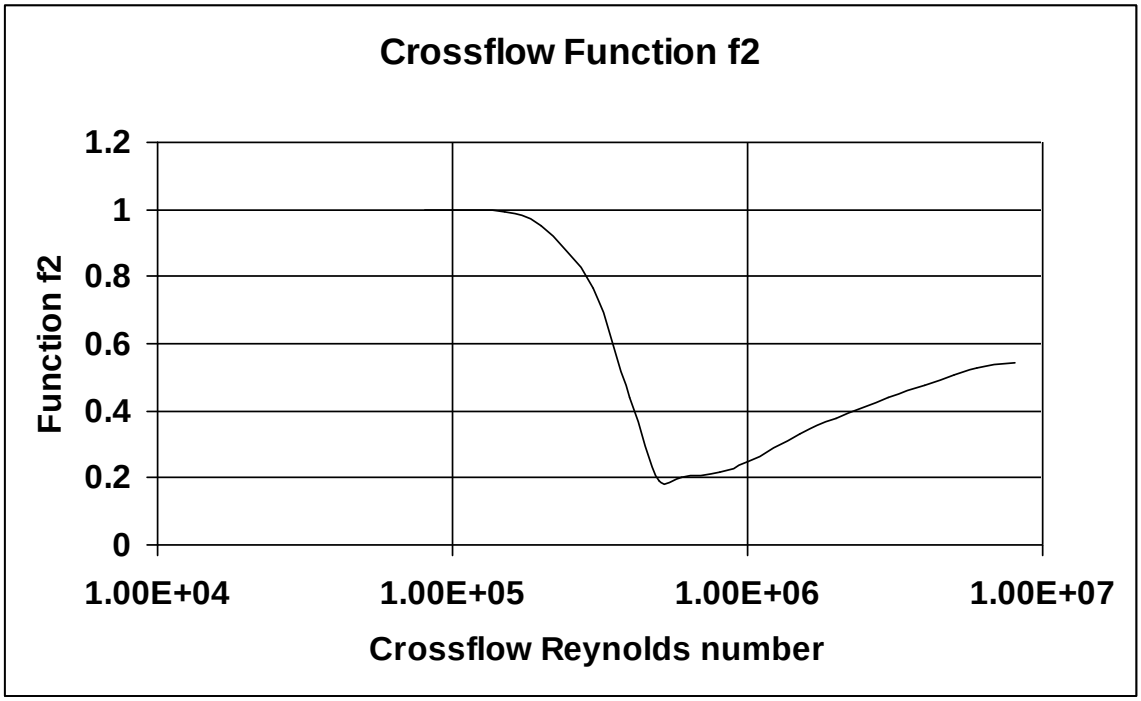
$$C_{A90} = 3\pi\eta C_{DN} S_{BROAD} / 8 S_{REF} \tag{22}$$

Crossflow Mach number	Function f_1
0.00	1.20
0.10	1.20
0.20	1.21
0.30	1.23
0.40	1.27

0.50	1.36
0.60	1.47
0.70	1.59
0.80	1.53
0.90	1.56
0.95	1.88
1.00	1.97
1.10	1.78
1.20	1.67
1.30	1.61
1.40	1.55
1.60	1.50
1.80	1.45
2.00	1.41
2.40	1.37
2.80	1.34
3.20	1.32
3.60	1.30
4.00	1.29
5.00	1.27
6.00	1.26
7.00	1.25
8.00	1.25

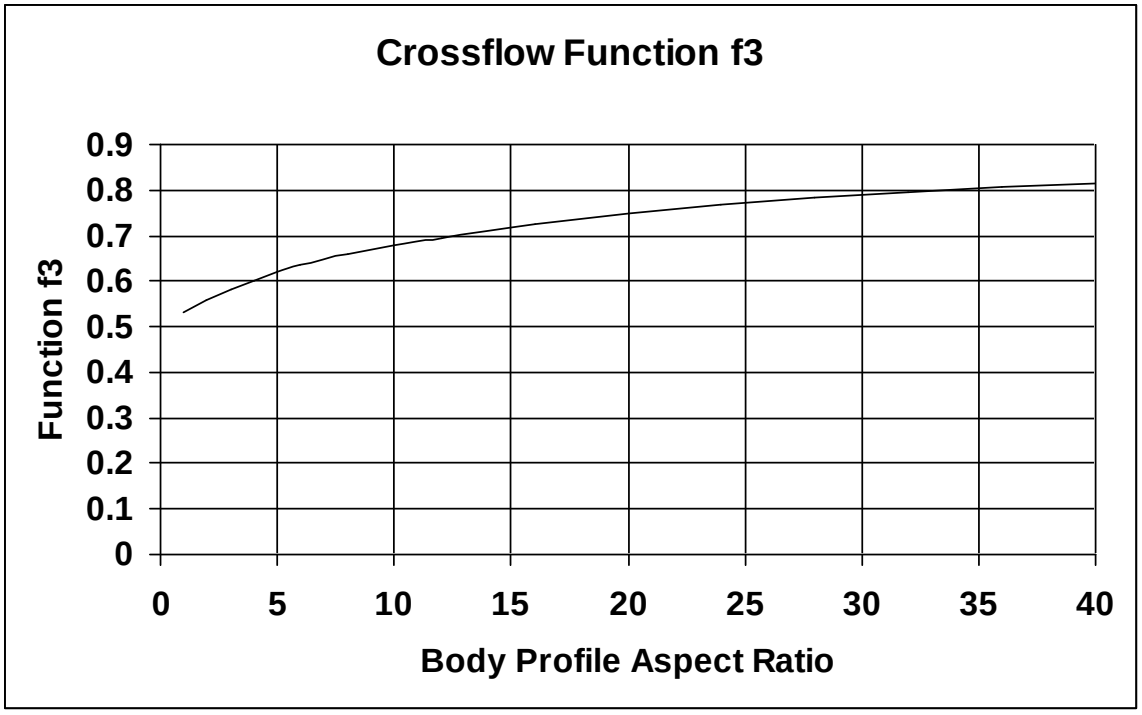
Crossflow Reynolds number	Function f_2
8.0E+04	1.000
1.0E+05	1.000
1.5E+05	0.992
2.0E+05	0.950
3.0E+05	0.767
4.0E+05	0.442
5.0E+05	0.192
6.0E+05	0.200
8.0E+05	0.217
1.0E+06	0.246
1.5E+06	0.333
2.0E+06	0.377
3.0E+06	0.442
4.0E+06	0.475
5.0E+06	0.508
6.0E+06	0.525
8.0E+06	0.542
1.0E+07	0.542

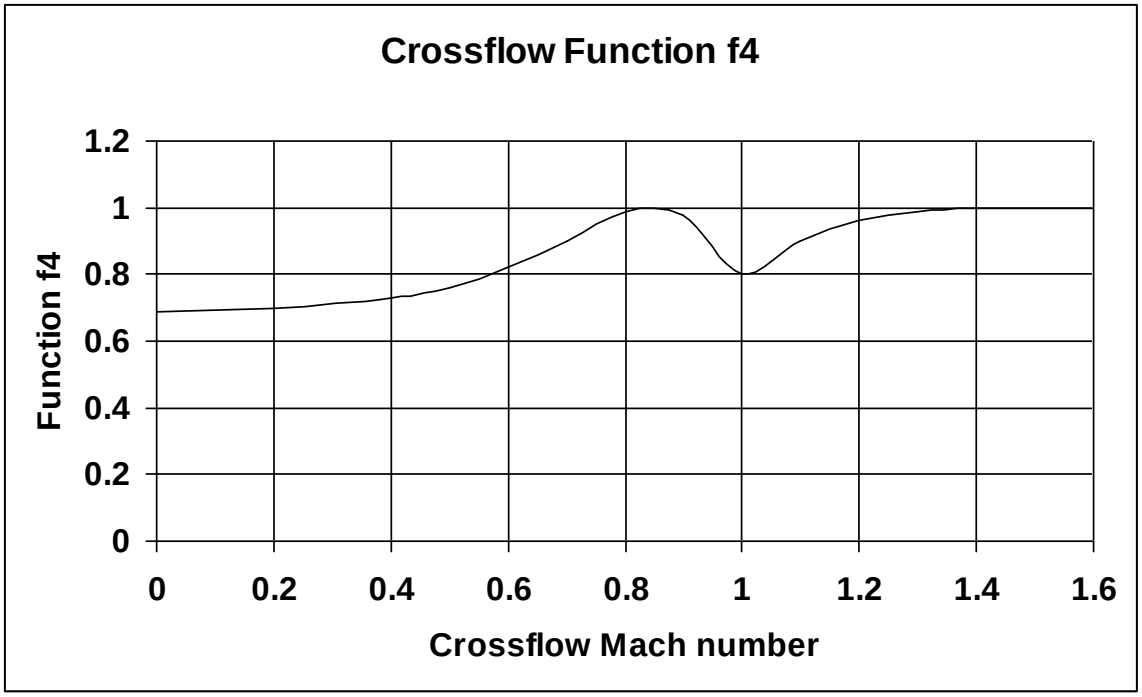

Crossflow function f1
Function f1
2
1.5
1
0.5
0
0
1
2
3
4
5
6
7
8
Crossflow Mach number


Crossflow Function f2
Function f2
1.2
1
0.8
0.6
0.4
0.2
0
1.00E+04
1.00E+05
1.00E+06
1.00E+07
Crossflow Reynolds number

Broadside aspect ratio	Function f_3
1.0	0.530
2.0	0.560
3.0	0.580
4.0	0.600
5.0	0.620
6.0	0.635
8.0	0.660
10.0	0.680
12.0	0.695
16.0	0.725
20.0	0.750
24.0	0.770
28.0	0.785
32.0	0.795
36.0	0.805
40.0	0.815

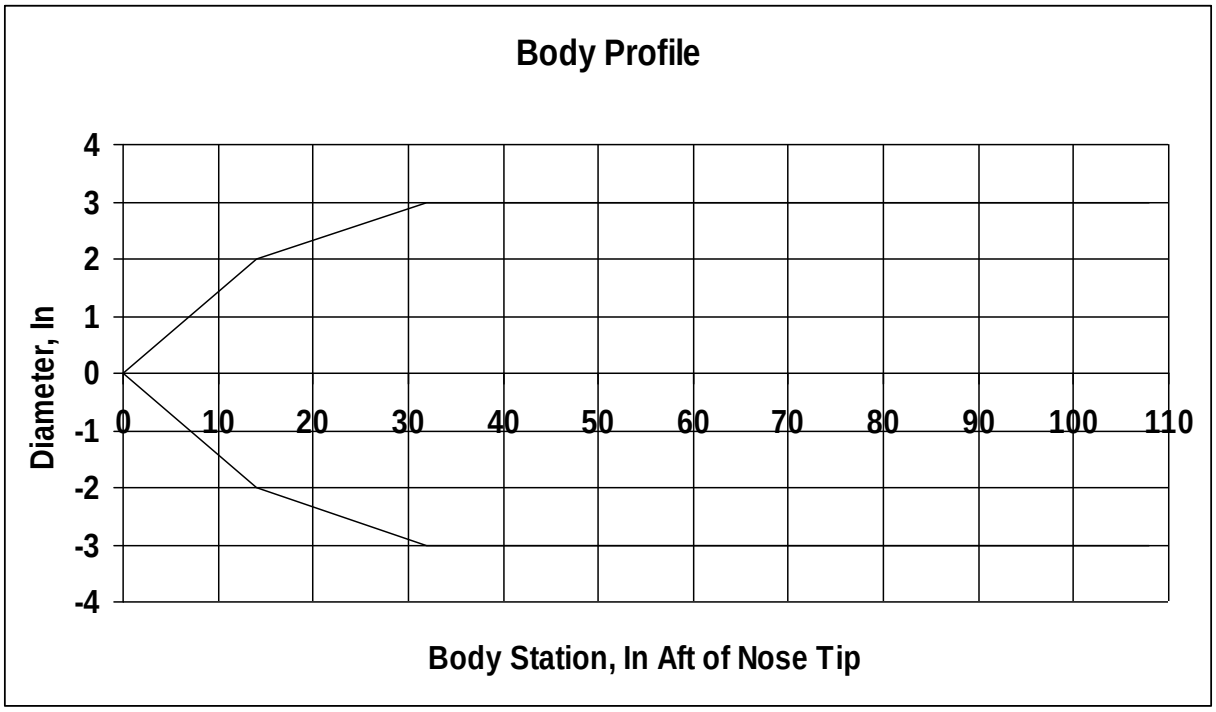
Crossflow Mach number	Function f_4
0.0	0.69
0.2	0.70
0.4	0.73
0.5	0.76
0.6	0.82
0.7	0.90
0.8	0.99
0.9	0.98
1.0	0.80
1.1	0.90
1.2	0.96
1.3	0.99
1.4	1.00
1.6	1.00

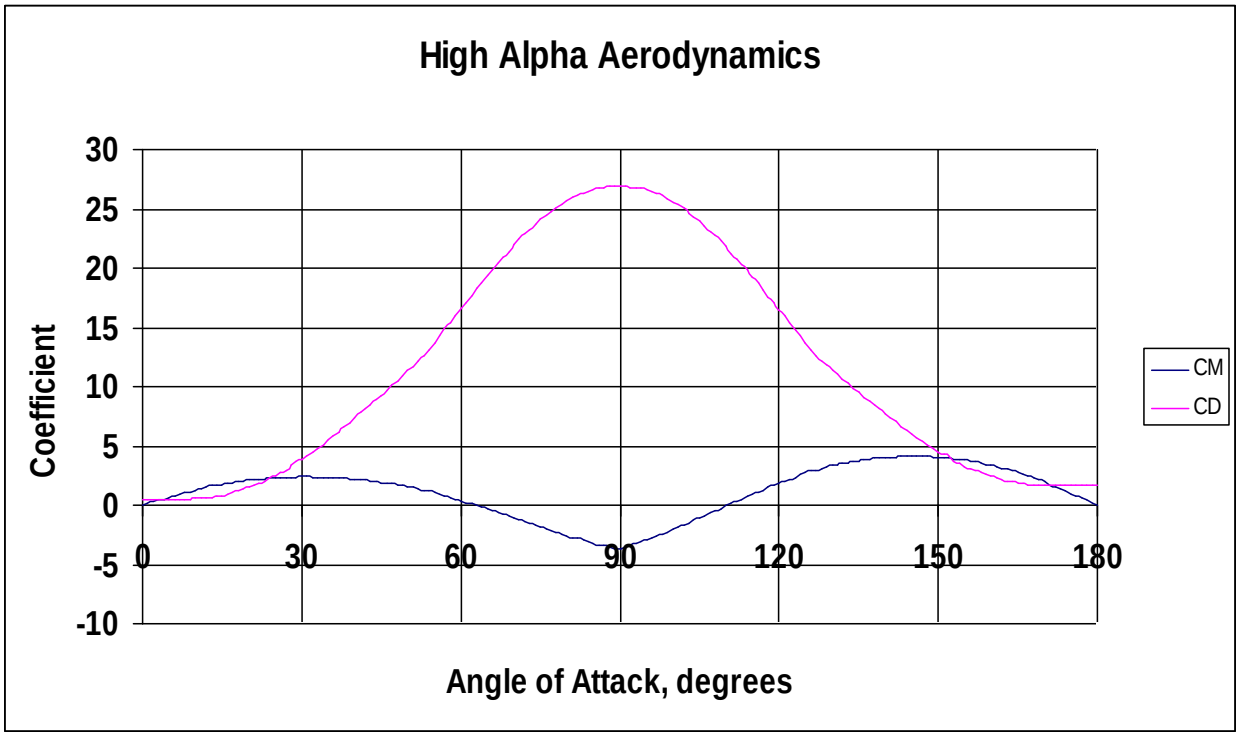




Typical Results

To better see how this works, consider a body with a biconic nose as sketched below flying at Mach number 2 with its center of gravity located 59 inches aft of the nose tip.





The figure above shows that any trim angle of attack between 45°and 135°will give significant aerodynamic deceleration during reentry. Two possible trim points in the figure are at a = 61.428° and 180°. The former would provide acceptable capability, but not the latter. Care must be taken to preclude this mode-it's very bad news.

Finally, the pitching moment curve is very sensitive to CG location. An inch either way on this 9 foot long body makes a significant change in the trim condition.

References

1. L. H. Jorgensen, "Prediction of Static Aerodynamic Characteristics for Slender Bodies Alone and with Lifting Surfaces to Very High Angles of Attack", N.A.S.A. TR R-474, September 1977
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